

Quantum Model and Periodic Trends Review Questions

1. Sketch 1s, 2s and 2p orbitals using the same scale for each. (ie. Need to see relative sizes of orbitals)
2. Write full electron configurations for the elements identified by these atomic numbers:
 - a) 9
 - b) 18
 - c) 21
3. Indicate which of these orbital designations is incorrect.
 - a) 4s
 - b) 1p
 - c) 2d
 - d) 3d
4. How many electrons are in the second energy level in an atom of these elements?
 - a) nitrogen
 - b) phosphorous
5. What is the maximum number of electrons that can go into each of the following sublevels?

a) 2s	e) 4p
b) 3p	f) 5s
c) 4s	g) 5f
d) 3d	
6. Give the symbol for the atom whose electron configuration corresponds to each of the following electron configurations:
 - a) $1s^2 2s^2 2p^6 3s^2 3p^6$
 - b) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^6$
 - c) $[\text{Ar}] 4s^2 3d^{10} 4p^3$

7. How many paired electrons are there in an atom of each of these elements?
- a) helium
 - b) boron
 - c) sodium
 - d) oxygen
8. Discuss the similarities and differences between:
- a) a 1s and 2s orbital
 - b) a $2p_x$ and a $2p_y$ orbital
9. List all the possible subshells and orbitals associated with $n = 3$.
10. Write the ground state electron configurations for the following elements and the orbital diagrams for each. Use the short form configuration for those that are allowed to be shortened.
- B, V, Sc^{2+}
11. What are the allowed values for the secondary quantum number and magnetic quantum number when the principal quantum number is 4?
12. What is wrong with each set of quantum numbers? Change one number to give an allowed set.
- a) $n = 0, \ell = 1, m_\ell = 0$
 - b) $n = 3, \ell = 2, m_\ell = -3$
13. What is the greatest number of electrons possible in each of the following energy levels?
- a) $n = 1$
 - b) $n = 3$

Textbook Questions:

Page 193 → Q's 10 - 12

Page 199 - 201 → Q's 7 - 13; 21, 22, 24, 27, 28, 30, 48

Page 204 - 205 → Q's 7, 9, 10, 18, 20

Answers - Quantum Model and Periodic Trends Review Questions

2. a) $1s^2 2s^2 2p^5$ b) $1s^2 2s^2 2p^6 3s^2 3p^6$ c) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$

3. b) and c) are incorrect

4. a) $N = 5$ b) $P = 8$

5. a) 2 b) 6 c) 2 d) 10 e) 6 f) 2 g) 14

6. a) Ar b) Ru c) As

7. a) 2 b) 4 c) 10 d) 6

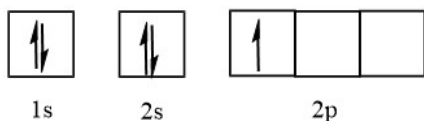
8. a) 1s is smaller volume than 2s; both are spherical orbitals

b) $2p_x$ is oriented along the x-axis, $2p_y$ is oriented along the y-axis; both are peanut shaped and have the same volume.

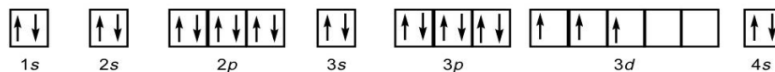
9. Subshells = 3s, 3p, 3d

Orbitals = 3s, $3p_x$, $3p_y$, $3p_z$, and 5 x 3d orbitals in different orientations

10. B: $1s^2 2s^2 2p^1$



V: $[Ar] 4s^2 3d^3$



Sc^{2+} : $[Ar] 3d^1$

(Loses 2 electrons from 4s orbital, leaving 1 electron in 3d)

11. $n = 4$

$\ell = (0 \text{ to } n - 1) = 0, 1, 2, 3$

$m_\ell = -\ell \text{ to } +\ell = -3, -2, -1, 0, 1, 2, 3;$

12. a) $n \neq 0$; change n to 2 or higher
b) $m_\ell \neq -3$; change m_ℓ to -2, -1, 0, 1, or 2
13. Number of electrons = $2n^2$
a) number of electrons = 2
b) number of electrons = 18

Textbook Questions:

Some of the answers to the textbook questions are at the back of the textbook, pg's 757-759.